

Rizin / radare2 Reverse Engineering Cheat Sheet

Rizin / radare2 Quick Cheat Sheet

- Open binary: rizin chall / r2 chall
- Analyze: aaa (full), aa (quick), aaaa (deep)
- List functions: afl
- Seek to function: s sym.main
- Disassemble function: pdf
- Strings: izz
- Cross references: axt @
- Visual graph mode: VV
- Decompiler (Rizin): pdg

Debugging

- Start debugger: rizin -d chall
- Set breakpoint: db sym.imp.strcmp
- Continue: dc
- Step: ds
- Inspect memory: px 64 @ rdi
- Print string: psz @ rdi
- Registers: dr
- Backtrace: dbt

Patching

- Assemble patch: wa je 0x401300
- Write and quit: wq

Common CTF Workflow

1. rz-bin -l chall (imports/checksec)
2. rizin chall
3. aaa
4. izz (find interesting strings)
5. axt on target strings
6. pdf on xref functions
7. pdg for pseudocode
8. Debug with breakpoints

- 9. Reimplement logic in Python

Useful Patterns

- Fisher-Yates: `rand() % (i+1) + swap`
- XOR loops
- Rolling hashes
- Base64 encode/decode chains
- CRC checks
- VM dispatchers
- Stack strings